Civil Excavation & Underground Utility JSA / JHA

Job Safety Analysis (JSA / JHA) - Deep Trenching & Utilities

Task Scope: Mechanical & Manual Excavation to -2.4m Depth - Permit Required: Excavation PTW

Step 1: Underground Service Scan
Potential Hazards: Strike on energized HV cable or pressurized pipe
Preventive Controls: As-built drawing check + calibrated CAT4 scan + mark route
Action By: Surveyor / HSE [HOLD]
PTW Gate

Step 2: Trial Pit Hand Digging
Potential Hazards: Mechanical bucket contact with buried conduit
Preventive Controls: Insulated hand tools within 500mm of marked service line
Action By: Civil Foreman
Verified

Step 3: Trench Battering & Shoring
Potential Hazards: Sidewall cave-in / worker entrapment (>1.2m depth)
Preventive Controls: Install hydraulic trench box or 45- benching + 1m spoil clearance
Action By: Site Engineer
Engineered

Step 4: Access & Gas Monitoring
Potential Hazards: Oxygen deficiency / H2S accumulation in deep cut
Preventive Controls: Multi-gas detector pre-entry check + secured ladder every 7.5m
Action By: Authorized Gas Tester
Logged

Step Sequence - Preparation & Survey

Each step is briefed to the crew before the shift starts; the JSA is re-validated if conditions change.

Review Drawings & Permits
Detail: Utility drawings may be incomplete or outdated
Control / Requirement: Permit-to-dig issued, drawings briefed at TBT
Verification: Permit copy

Mark Out Excavation
Detail: Wrong location - service strike
Control / Requirement: Survey pegs + spray lines verified with supervisor
Verification: Walk-round

Set Up Work Zone
Detail: Public / plant interface
Control / Requirement: Hard barriers + signage before breaking ground
Verification: Checklist

Brief the Crew (TBT)
Detail: Unbriefed worker enters unsafe zone
Control / Requirement: Task-specific TBT signed by all crew
Verification: TBT record

Step Sequence - Utility Scan & Trial Pits

Each step is briefed to the crew before the shift starts; the JSA is re-validated if conditions change.

CAT & Genny Scan
Detail: Undetected live cable
Control / Requirement: Calibrated scanner, trained operator, grid pattern scan [HOLD POINT]
Verification: Scan record

Hand-Dug Trial Pits
Detail: Tool strike on service
Control / Requirement: Insulated tools, hand-dig within 500 mm of marks
Verification: Supervisor check

Expose & Support Services
Detail: Unsupported service sagging/damage
Control / Requirement: Timber supports, service owner notified
Verification: Photo record

Re-Scan After Exposure
Detail: Deeper unmarked services
Control / Requirement: Second scan pass before machine dig
Verification: Scan record

Step Sequence - Machine Excavation

Each step is briefed to the crew before the shift starts; the JSA is re-validated if conditions change.

Position Excavator
Detail: Plant overturn / edge surcharge
Control / Requirement: Firm level pad, spoil - 1 m from edge or per design
Verification: Operator check

Dig in Layers
Detail: Undermining trench wall
Control / Requirement: Shallow passes, banksman watching walls
Verification: Visual

Spoil Management
Detail: Spoil rollback into trench
Control / Requirement: Toe boards / stop blocks, haul route segregated
Verification: Daily check

Dewatering
Detail: Water softening trench base
Control / Requirement: Sump + pump, discharge to approved point
Verification: Inspection

Step Sequence - Shoring & Access

Each step is briefed to the crew before the shift starts; the JSA is re-validated if conditions change.

Install Support System
Detail: Collapse during installation
Control / Requirement: Install from safe position, designed system only [HOLD POINT]
Verification: Competent person

Ladder / Ramp Access
Detail: Fall entering trench
Control / Requirement: Secured ladders extending 1 m above lip, every 15 m
Verification: Daily check

Edge Protection
Detail: Person or material falls in
Control / Requirement: Guardrails or hard barriers at all open edges
Verification: Walk-round

Daily Inspection
Detail: Ground movement after rain/weekend
Control / Requirement: Competent-person inspection before each shift + after rain [HOLD POINT]
Verification: Inspection form

Step Sequence - Work Inside the Excavation

Each step is briefed to the crew before the shift starts; the JSA is re-validated if conditions change.

Pipe / Duct Laying
Detail: Load swing over workers
Control / Requirement: No workers under suspended loads, tag lines used
Verification: Lift plan

Compaction Works
Detail: HAVS / plant interface
Control / Requirement: HAVS log, pedestrian exclusion while compacting
Verification: Log

Housekeeping in Trench
Detail: Trips in confined trench floor
Control / Requirement: Materials stacked clear of access route
Verification: Supervisor check

Atmosphere Awareness
Detail: Gas accumulation in deep trench
Control / Requirement: Test atmosphere where depth/source requires
Verification: Gas test record

Residual Risk Summary & ALARP Statement

All activities are reduced to Low/Medium residual risk once the stated controls are verified in place. No activity may start on initial risk alone.

High (15-25) initial
Count of Activities: Assessed per activity matrix
Action Rule: Controls + hold points mandatory before start
Authority: Site HSE authority

Medium (8-12) residual
Count of Activities: Monitor at supervision level
Action Rule: Daily TBT reinforcement + spot checks
Authority: Supervisor

Low (1-6) residual
Count of Activities: Majority of activities after controls
Action Rule: Proceed under permit and supervision
Authority: Permit issuer

ALARP Statement
Count of Activities: Further risk reduction considered for every High initial item
Action Rule: Grossly disproportionate measures documented, not silently dropped
Authority: HSE Officer

Hold Point Register

Work must stop at each hold point until the named authority verifies and releases it. Hold points are never self-released.

HP-01 Permit Verification
Where It Applies: Start of each high-risk activity
Released By: Permit issuer
Evidence: Signed permit at workface

HP-02 Isolation / LOTO
Where It Applies: Energy isolation before work
Released By: Authorised isolator
Evidence: Lock register + prove-dead test

HP-03 Pre-Start Inspection
Where It Applies: Plant, scaffold, lifting gear
Released By: Competent inspector
Evidence: Tag / inspection record

HP-04 Authority Sign-Off
Where It Applies: Document approval before execution
Released By: Client/site HSE authority
Evidence: Signed approval block

Review, Approval & Revision Record

This document is a professional draft. Field use requires review and formal approval by the client/site HSE authority.

Prepared By
Name / Organisation: SafetyWithAthar (remote)
Responsibility: Draft preparation from supplied work-pack inputs
Status: Rev. 0 issued

Reviewed By
Name / Organisation: [SITE INPUT REQUIRED]
Responsibility: Site HSE review against actual conditions
Status: [HOLD POINT]

Approved By
Name / Organisation: [SITE INPUT REQUIRED]
Responsibility: Client / site HSE authority approval
Status: [HOLD POINT]

Revision History
Name / Organisation: Rev. 0 - first issue for review
Responsibility: Superseded revisions withdrawn from field use
Status: Controlled